Doyun (Logan) Kim

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Education

University of Washington Seattle, WA
Bachelor of Science in Electrical and Computer Engineering – GPA: 3.59 /4.0 *Exp. Grad. Dec 2026*

Skills

Programming: Python, C, C++, Java, MATLAB, SystemVerilog, Bash

Tools: Fusion 360, KiCAD, SolidWorks, VSCode, Arduino IDE, git, ROS2, Docker, ModelSim, MATLAB Simulink, Gazebo, Microsoft Office Suite

Technical Skills: 3D printing, CAD Modeling, Electronic Designing, Soldering, Linux (Ubuntu)

Projects

LAIKA, the Quadrupedal Robot

- Designed and developed a quadrupedal robot by modeling 12 body parts in Fusion 360 and 3D printing them.
- Conducted power analysis of the robot to achieve the operation time of ~22 minutes at the typical current draw of 13.25 A.
- Tested the functionality of electric parts like servo motors and sensors with custom Python scripts.
- Analyzed the DH parameters of the robot for forward kinematics and wrote the gait control logic via inverse kinematics.
- Simulated kinematics and real-world navigation of the robot in MATLAB Simulink.
- Implemented ROS-base framework with Ubuntu-booted Raspberry Pi for efficient system organization.

Face Tracking Turret

- Designed a 3 DOF face-tracking turret capable of stable detection and tracking of faces at ~30 FPS controlled with ESP32.
- Configured serial connection between the Python agent and ESP32 with PySerial to achieve real-time update of servos.
- Implemented face detection with OpenCV's Haar Cascade classifier and precise tracking in servos with PD control.
- Optimized joint movement using deadband and rate-limiting to prevent servo overshoots, ensuring smooth turret motion.

Experience

UW Solar Vehicle Team Sep 2025 – Present
Electrical Engineer *Seattle, WA*

- Designed power distribution board that distributes 24 V battery inputs to 8 different electrical subsystems used in the solar car.
- Prototyped the electrical boards to verify the functionality of the board design and to document their stress limits.
- Conducted layout of the schematics into a PCB with 4 conductive layers and traces that can handle corresponding load currents.
- Implemented a voltage monitoring system for the supplementary battery pack for potential unsafe operating conditions.

L'Space NPWEE Academy May 2024 – July 2024
Research Engineer *Remote*

- Trained in the NPWEE Academy to develop potential solutions to technological pain points in NASA's space missions.
- Submitted 2 proposals complying with the eligibility guidelines for NASA's New Technological Report system (NTRs).
- Researched chemical properties of Bismuth Telluride substances for thermoelectric paints that convert waste heat into electricity.

US 8th Army Korean Augmentation to the United States Army June 2022 – October 2023
Signals Intelligence Voice Interceptor(35P) *Pyeongtaek, South Korea*

- Led a team of 7 SIGINT mission specialists composed of 3 Korean Army soldiers and 4 U.S. Army soldiers.
- Established an efficient communication channel between two intelligence commands across the ROK Army and the U.S. Army.
- Acquired security clearance and disseminated Secret-level intelligence by intercepting and transcribing foreign communications
- Operated and maintained the signal processing software that handle communication traffic with different radio frequencies.